

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Dept. of Metallurgical and Materials Engineering

2. Subject Name: Introductory Scientific Computing for Materials Science

L-T-P: 3-0-0

Credits: 3

3. Pre-requisites: None

4. Syllabus and reference books:

Syllabus:

(1) Introduction: Scope of scientific computing; Difference between technical computing and other algorithmic schemes; Scientific computing in materials science; Role of numerical methods in scientific algorithms; Applications of AI in scientific computing.

(2) Floating point representation: Computer representation of a floating point number; Machine epsilon; IEEE protocol for floating points; Hidden bit normalization; Algebraic properties of floating point numbers.

**Case study: Critical system failure*

(3) Solutions of nonlinear equations: Bisection method and its convergence; Newton-Raphson method and its convergence criteria; Secant method and its convergence; Termination criteria for iterative solvers; System of nonlinear equations and its solution using multi-variable Newton-Raphson and Broyden's methods.

**Case study: Finding tolerance limit for creep behavior*

(4) Simulating stochasticity in a digital computer: Methods of generating random numbers; Modulo generator and theorems regarding its periodicity; Shift-register generator; Monte Carlo integration; Generation of non-uniform random numbers; Box-Muller method and Marsaglia polar form.

**Case study: Random walk model of diffusion*

(5) Solutions of linear systems: Solving simultaneous equations through LU factorization; Gauss' algorithm for LU decomposition; Criterion for existence of LU factors; Special cases of coefficient matrices.

**Case study: Finding lattice parameters of constituent elements of high-entropy alloys assuming Vegard's law*

(6) Numerical solutions of initial value problems: Vector representation of a higher order ODE; Numerical solution with Euler's method; Implicit and explicit schemes; Crank-Nicolson method; Taylor's method; Runge-Kutta method and derivation of RK2; RK4 method.

**Case study: Solution of 1-D diffusion equation*

(7) High performance scientific computing: Von Neumann architecture; Pipelining and memory bank; Cache oblivious programming; Parallel architecture; Message passing topologies; Amadahl's law.

**The prescribed case studies are merely representative. An instructor may choose different demonstrations as per his/her convenience and consideration.*

Reference Books:

- (1) James F. Epperson, *An Introduction to Numerical Methods and Analysis*, Wiley.
- (2) J. M. Thijssen, *Computational Physics*, Cambridge University Press.

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Introduction	2
2.	Floating point representation	4
3.	Solutions of nonlinear equations	6
4.	Simulating stochasticity in a digital computer	7
5.	Solutions of linear systems	5
6.	Numerical solutions of initial value problems	7
7.	High performance scientific computing	5
Total number of hours		36